Melissa Gibney

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EDUCATION

Carnegie Mellon University **Pittsburgh, PA**
Master of Human Computer Interaction (MHCI) *August 2024 – August 2025*

- GPA: 4.03/4.00

University of Rochester **Rochester, NY**
Bachelor of Arts in Computer Science *August 2019 – May 2024*
Bachelor of Science in Audio and Music Engineering *August 2019 – May 2024*

- GPA: 3.91/4.00 (Dean's List, Cum Laude)

WORK EXPERIENCE

Carnegie Mellon University **Pittsburgh, PA**
CyLab Extern *September 2025 – January 2026*

- Implemented picoCTF's website redesign with a homepage, profile page, and learning paths using React. This work was a continuation of the picoCTF Capstone Project found below.

University of Rochester **Rochester, NY**
Undergraduate Teaching Assistant *August 2021 – December 2022, August 2023 – May 2024*

- Helped students understand the concepts involving audio acoustics, Fourier transforms, and MATLAB code.
- Advised students on how they should proceed during assignments involving Faust, JUCE, and C++.

Emagination STEM Camps **Lancaster, PA**
Program Coordinator *June 2023 – August 2023*

- Taught students (ages 8 to 17) to use C#, Python, and Unreal Engine 5 in various week-long courses.
- Organized activities and ensured that students remained safe during their stay at the camp.

General Dynamics Mission Systems **Dedham, MA**
Software Engineering Intern *June 2022 – August 2022*

- Updated Agent Starling (GEM One testing software) to emulate 2000 devices and run on a Tomcat server.
- Documented instructions for configuring and running Agent Starling on a virtual machine.

SELECTED PROJECTS

LLM-Powered Discord Chatbot: Created a chatbot that can converse with users in Discord and respond while staying in-character using an LLM, Node.js, and Python.
picoCTF Capstone Project: Researched ways to improve user engagement and retention on the picoCTF cybersecurity platform as a part of a five-person team. Implemented resulting designs as React components with Typescript—using Storybook for documentation and Docker for containerization and shareability.
Character Creator React App: Designed a web application that allows users to customize their own character using premade pixel art assets. Each design can then be exported as an SVG or PNG.
Flute Synthesizer: Coded a synthesized flute in Max/MSP based upon research by Professor Perry R. Cook.
Robotic Drum Machine: Worked in a team of three to design and prototype a robotic drum machine that actuates solenoids according to a sequence input by the user. The user could change the tempo and select which solenoids to actuate.

SKILLS

- Programming Languages, Libraries, and Frameworks: JavaScript, Typescript, React, HTML, CSS, Python, Java, C, C++, C#, SQL, Swift, Flask, MATLAB, R, Faust, JUCE, and Max
- Tools: Git, GitHub, Docker, AWS, Figma, FigJam, Miro, Jira